DEL MAR
EST. SEAFOODS 1988

NOTES FROM QINGDAO · 青岛会谈备忘

PLATE FREEZER PROJECT NOTES

平板冻结项目备忘

What we talked about and what we're after, so Wolfking can get started. English version; the Chinese version is a separate file.

DEL MAR SEAFOODS · 30 OCTOBER 2026 · FOR WOLFKING AND APCCO

A note on these notes: they were put together with AI from the meeting transcripts and the email chains, and I don't have the refrigeration or regulatory expertise to check every line. If anything here conflicts with what Wolfking or APCCO know, they're right and this is wrong.

WHO'S DOING WHAT

- **Del Mar (Joe)** — the customer. Picks the block size, approves the layout and the final design.
- **Wolfking** — designs and builds four vertical plate freezers with hydraulics and controls, plus the ammonia package to run them. Supplies drawings and the paperwork the permits need, and two technicians to help APCCO install and start up.
- **APCCO** — our refrigeration contractor in California. Checks the design against California rules, stamps and files the permits, installs everything and services it afterwards. They don't build freezers. They'll talk directly to your engineers and ask a lot of questions; that's just the permit process.
- One email thread, all three of us copied.

> **Sean, about the Poland story.** Regulations are on us, not you. Our job (with APCCO) is to work out what's needed here and order the right thing; yours is to build machines that match the spec and work. APCCO will send you the list of certificates and documents we need before anything is built; you tell us which you can provide. Anything missed on the regulatory side after that is our mistake to fix, not yours.

WHAT WE'RE MAKING

- **Whole California squid** in 10 kg blocks, a test for the China market. Squid season starts in May, so we want to be running by May 2027.
- **Ground salmon racks** (frames left after filleting, ground to a paste) for a pet-food customer, year-round. That contract pays for the machines.

45 t	2 × 10 h	4	140
a day (50 short tons)	shifts + 4 h cleaning	vertical plate freezers	blocks of 10 kg per batch, per freezer (the wish)

Our math: 90 min freeze + 30 min unload/reload = a batch every 2 hours, 4 per shift, 8 a day. 45 t ÷ 8 ÷ 4 freezers ≈ 1.4 t, or 140 blocks, per freezer per batch. A bit lower is okay.

Today we blast-freeze squid in cartons on racks, a lot of hand labor at $30 an hour. These four are a test. If it works we add more of the same design, and later automate loading, unloading, bagging and palletizing.

THE BLOCK

- 10 kg, about 12 litres (12,000,000 mm^3); squid is roughly 20% lighter than water.

- **75 mm thick.** Fixed.
- Face around 535 × 300 mm. We're flexible on the exact face: pick what fits your plates and still gives 12 L. It becomes our standard block for every freezer after this, so let's agree it before plates are ordered.
- Two dividers per station, three blocks across. Check that works with your plate height.
- 90 minutes is our guess for 75 mm. Give us the real number for whole squid and for salmon paste.
- **Drainage matters.** Whole squid dumps water and ink when loaded; the station bottom has to let it out so the customer gets squid, not ice. The paste must not leak out. We need a bottom that handles both. Photos or samples on request.

THE FREEZERS

- Four identical vertical plate freezers, indoors, top-loaded. Hydraulic open/close and lift, hot-gas release, blocks out one side onto a table or conveyor.
- Loading manual or semi-automatic to start (conveyor with gates for squid, hose and pump for paste; your ideas welcome). Build in the interface for your automatic loading and unloading later: top clearance, mounting points, spare I/O.
- Blocks go into bags and are stacked on pallets; no boxes. Quote your partner's bagger as an option.
- Aluminum plates. Stainless or aluminum wherever product or wash water touches; nothing galvanized above the plates. Base frame: quote stainless and hot-dip galvanized with the difference. Salt water and chlorine washdown daily.
- Hydraulic unit: 460 V / 60 Hz, out of the cold and wet area, H1 food-grade oil, −35 °C at the cylinders. One per freezer or shared?
- Stainless ammonia-rated hoses, replaceable from outside. No copper or brass near the ammonia.
- E-stops, guarding at the open top, lockable disconnect. APCCO will check against US rules.
- English manuals, drawings, schematics, a two-year spares list, STEP models.

REFRIGERATION

- **Ammonia (R717).** Freon isn't an option in California at this size, even though it works for your Oregon customers; details in the California section. CO_2 would also pass if that's something you do.
- Wolfking supplies the whole package: compressors, condenser, vessels, pumps, valves, controls. APCCO does the site piping and wiring.
- **Containerized engine room, built as skid-mounted portable equipment:** on skids in the yard next to the building, hard-piped to the freezers inside, liftable and movable as one unit (lifting points, one electrical connection, flanged pipe connections). Joe and APCCO pick the spot in early November; send container dimensions and skid weight before then. Joe also asked about a small self-contained unit per freezer: realistic with ammonia? Cost?
- **Keep the ammonia charge as low as you can.** 500 lb (227 kg) is where California's extra paperwork starts. If we can't get under it, just tell us the number.

- APCCO's early estimate (before 140 blocks): about 45 TR at −30 °F (−34 °C) suction, pumped recirculation. Do your own load calc for the final batch size.
- Tell us: system type and recirculation ratio; suction and condensing temperatures; condenser type (evaporative or air-cooled; water and noise both matter); receiver or not; defrost method and what the freezers need for it; which valves come mounted, loose, or from APCCO, with brands.
- ASME U-stamp vessels, B31.5 piping, plates rated for discharge pressure (hot gas hits them). APCCO gives the numbers.

POWER

- 480 V, 3-phase, 60 Hz; motors nameplated 460 V / 60 Hz. No 380 V and no 220 V single-phase in the US.
- Controls on 120 VAC and 24 VDC.
- Panels to UL 508A (UL or ETL listed), NEMA 4X stainless where there's washdown, name-brand UL components (Schneider, ABB, Siemens, Eaton, Allen-Bradley).
- Motors NEMA Premium / IE3 (required for imports, even inside a machine). Size pumps and fans for 60 Hz speed.
- **Send a load list early:** kW, amps and starting current per freezer and per skid. We need it for the power company, which can take longer than building the machines.
- PLC and screens in English. Remote access is fine. Siemens or Allen-Bradley is easiest to service here.

THE SITE

- Watsonville, California, on Monterey Bay, about 150 km south of San Francisco. Mild coastal weather; APCCO will send condenser design temperatures.
- Send weights, footprints, floor loads and lifting points for the freezers and the skid.
- Ship to Oakland or Long Beach, truck to Watsonville. Keep the package within container or road-legal size.

CALIFORNIA THINGS TO KNOW WHILE YOU DESIGN

More detail here because it drives the design. APCCO owns all of it and gives the exact numbers.

REFRIGERANT: WHY NOT FREON

- California (CARB), since 2022: a new system over 50 lb of charge in a new facility needs a refrigerant with GWP under 150. R-404A is about 3,900, R-507 about 4,000, R-448A/R-449A about 1,390, even R-32 is 675. Under 150 leaves ammonia (0), CO_2 (1), or A2L blends like R-454C/R-455A (about 148), which we don't want at this size.
- Federal (EPA): process refrigeration with evaporators between −30 and −50 °C can't use GWP 700 or more from 1 January 2028. Our later freezers would land after that; we won't build on a refrigerant we can't repeat.

- Over 50 lb of any HFC also means California's refrigerant management program: registration, leak inspections, reporting. Ammonia and CO_2 are outside it.
- So: ammonia, as planned. CO_2 also passes.

AMMONIA RULES

- Design to ANSI/IIAR 2-2021; install, start up and inspect to IIAR 4, 5 and 6; plus the California Mechanical Code. APCCO works to these daily.
- On the drawings: high-side design pressure (typically 250 psig / 17.2 bar; plates and hot-gas piping carry it; APCCO confirms); relief valves on every vessel, piped to a safe point outdoors; a labelled emergency shutoff; pipe and valve labels (refrigerant, pressure level, direction); ammonia detectors in the freezer room and at the package, wired to alarm and shut down.
- Ammonia equipment inside a room makes it a "machinery room" (ventilation, detection, door controls, restricted access, no open flames). The outdoor package avoids most of that. Tell us whether your container is built as a machinery room (fans, detectors, door switches) or as open-air equipment.
- Paperwork by charge: any ammonia means a hazardous-materials plan with the county; 500 lb or more adds CalARP (risk management plan, hazard review, inspections); 10,000 lb adds federal PSM/RMP, which we'll never reach. Lowest charge, least paperwork, and we need the number early.
- Leave room for an eyewash and shower at the package.

VESSELS AND PIPING

- Vessels: ASME Section VIII Div. 1, U-stamp, National Board number. Cal/OSHA permits each ammonia vessel and wants weld radiography; APCCO says which and how much.
- Piping: ASME B31.5; welders to ASME IX; material certificates; pressure-test records for anything shipped pre-piped.
- No copper or brass in the ammonia circuit; hose data sheets; plate pressure rating and test pressure on paper.

ELECTRICAL

- California Electrical Code is the NEC plus California additions. Every item listed by a US lab (UL, ETL or CSA); the inspector reads the labels. Unlisted gear means a field evaluation: slow, expensive, sometimes fails.
- Panels to UL 508A with the SCCR on the label; NEMA 4X in washdown, 3R or 4 outdoors; NFPA 79 wiring and colours; English labels and drawings.
- A lockable disconnect within sight of each motor, and a main disconnect per freezer and per skid.
- Motors at DOE efficiency (NEMA Premium / IE3) with the efficiency on the nameplate; VFDs and soft starters listed too.

FOOD SAFETY

- FDA: seafood HACCP for the squid, animal-food rules for the paste. Cleanable, corrosion-resistant contact surfaces; no flaking galvanizing or paint over product; drains and slopes; NSF H1 lubricants where product could be touched.

WORKER SAFETY

- Cal/OSHA: guarded pinch points, E-stops, lockout points for electrical and hydraulic energy, English warning labels. Any loading platform needs guardrails, toe boards and proper stairs; include it in the design.

NOISE

- Limits at the property line, lower at night, and we run 20 hours a day. Give condenser and compressor noise (dB(A) at 10 m) with the proposal.

PERMITS, AND WHO FILES WHAT

- The refrigeration package is skid-mounted portable equipment and is handled as equipment, not construction. Keep it that way: lifting points, one electrical connection, flanged pipe connections, nothing cast into the ground.
- Work on the building (feeder to the skid, freezer room, wall penetrations) goes through normal permits: APCCO, on stamped drawings.
- Unaffected by how the package is mounted: the refrigerant rules (anything in one place over 12 months is stationary to CARB and EPA), the county's hazardous-materials plan and CalARP (quantity on site), Cal/OSHA vessel permits and ASME stamps, electrical listing, food safety, worker safety, noise.
- County (CUPA) and CalARP: Del Mar with APCCO. Cal/OSHA vessel permits: APCCO. PG&E service and food-plant registration: Del Mar. The county and PG&E paperwork can take longer than manufacturing, so drawings first.

> **Paperwork you'll be asked for.** Stampable drawings (layout, P&ID, one-line, schematics); design-pressure and relief-valve sizing calcs; ASME U-1/U-1A reports and National Board numbers per vessel; material certificates and welder qualifications; pressure-test records; UL 508A label and SCCR calc per panel; motor efficiency nameplates; weights and lifting points; ammonia charge calc; English O&M manuals. If you can't provide one, say so now and APCCO will find a way around it.

TIMING

- **Now:** these notes; Joe and APCCO pick the layout; Wolfking starts.
- **ASAP:** preliminary drawings, P&ID, load list, ammonia charge estimate, proposal. Permits start from the preliminary design.
- **Build:** about 100 days after design approval, per your estimate at the show.
- **Running by May 2027,** June at the latest. Joe told the pet-food customer six months.
- **2027–28:** automatic loading and unloading, bagging, palletizing, check-weighing.

WHAT WOULD HELP US MOST RIGHT NOW

1. Preliminary layout of one freezer: size, plate count, block-face proposal, loading height, discharge side, weight.
2. Freeze-time check for 75 mm whole squid and salmon paste.
3. Refrigeration concept: system type, P&ID, equipment list, ammonia charge, container dimensions.
4. Electrical load list and one-line.
5. Vessel list with ASME status, and which certificates you can supply.
6. Proposal: scope split with APCCO, price, delivery, terms, spares, technician terms.
7. STEP models when you have them.
8. A couple of US references we can call (Oregon, LA).
9. A ballpark for phase-2 automation and the bagger, so we design the interfaces now.

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